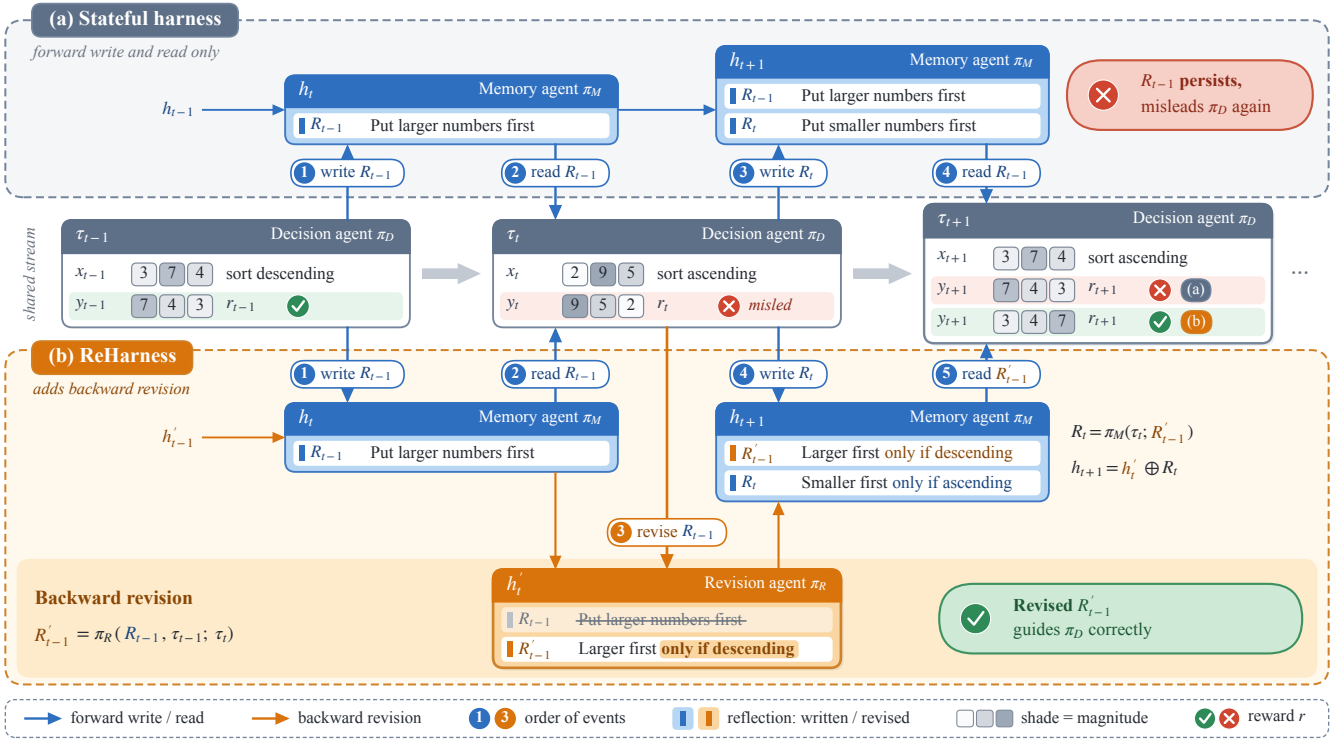

(a) Stateful harness
forward write and read only
h_{t-1}
h_t Memory agent π_M
R_{t-1} Put larger numbers first
h_{t+1} Memory agent π_M
R_{t-1} Put larger numbers first
R_t Put smaller numbers first
R_{t-1} persists, misleads π_D again
1 write R_{t-1}
2 read R_{t-1}
3 write R_t
4 read R_{t-1}
shared stream
τ_{t-1} Decision agent π_D
x_{t-1} 3 7 4 sort descending
y_{t-1} 7 4 3 r_{t-1}
τ_t Decision agent π_D
x_t 2 9 5 sort ascending
y_t 9 5 2 r_t misled
τ_{t+1} Decision agent π_D
x_{t+1} 3 7 4 sort ascending
y_{t+1} 7 4 3 r_{t+1} (a)
y_{t+1} 3 4 7 r_{t+1} (b)
...
(b) ReHarness
adds backward revision
1 write R_{t-1}
2 read R_{t-1}
4 write R_t
5 read R'_{t-1}
h'_{t-1}
h_t Memory agent π_M
R_{t-1} Put larger numbers first
h_{t+1} Memory agent π_M
R'_{t-1} Larger first only if descending
R_t Smaller first only if ascending
$R_t = \pi_M(\tau_t; R'_{t-1})$
$h_{t+1} = h'_t \oplus R_t$
3 revise R_{t-1}
Backward revision
$R'_{t-1} = \pi_R(R_{t-1}, \tau_{t-1}; \tau_t)$
h'_t Revision agent π_R
R_{t-1} Put larger numbers first
R'_{t-1} Larger first only if descending
Revised R'_{t-1} guides π_D correctly
forward write / read
backward revision
order of events
reflection: written / revised
shade = magnitude
reward r